Project Documentation

Project Title

Week 2 – Data Preprocessing and Cleaning Pipeline

Introduction

Data preprocessing is one of the most important steps in machine learning. Real-world datasets often contain missing values, duplicate records, inconsistent information, and categorical data that cannot be directly used by machine learning algorithms.

The purpose of this project is to clean and prepare the dataset before applying machine learning models. A properly cleaned dataset improves model accuracy and overall performance.

Objective

The main objectives of this project are:

- Load and understand the dataset
- Identify and handle missing values
- Remove duplicate records
- Convert categorical data into numerical form
- Apply feature scaling to numerical columns
- Save the cleaned dataset for future machine learning tasks

Dataset Information

Dataset Name: Teen_Mental_Health_Dataset.csv

The dataset contains information related to teen mental health. It includes both numerical and categorical features.

Technologies and Libraries Used

Tool / Library	Purpose
Python	Programming language
Pandas	Data manipulation and analysis
NumPy	Numerical operations
Scikit-learn	Encoding and feature scaling

Data Preprocessing Steps

Step 1: Load the Dataset

The dataset was loaded using the pandas library.

Code

```
import pandas as pd

# Load dataset
file_path = 'Teen_Mental_Health_Dataset.csv'
df = pd.read_csv(file_path)

# Display first rows
print(df.head())
```

Purpose

- Read the CSV dataset
- Display initial rows
- Understand dataset structure

Step 2: Dataset Information

The dataset structure, column names, and data types were checked.

Code

```
print(df.shape)
```

```
print(df.columns)
print(df.dtypes)
print(df.info())
```

Purpose

- Identify total rows and columns
- Understand feature types
- Detect numerical and categorical columns

Step 3: Identify Missing Values

Missing values were identified using pandas functions.

Code

```
print(df.isnull().sum())
```

Purpose

- Detect incomplete data
- Find columns requiring cleaning

Step 4: Handle Missing Values

Numerical missing values were filled using mean, while categorical values were filled using mode.

Code

```
# Numerical and categorical columns
numerical_cols = df.select_dtypes(include=['int64', 'float64']).columns
categorical_cols = df.select_dtypes(include=['object']).columns

# Fill numerical values
for col in numerical_cols:
df[col].fillna(df[col].mean(), inplace=True)

# Fill categorical values
```

```
for col in categorical_cols:
df[col].fillna(df[col].mode()[0], inplace=True)
```

Purpose

- Prevent data loss
- Replace incomplete values with meaningful values
- Improve dataset quality

Step 5: Remove Duplicate Records

Duplicate rows were identified and removed.

Code

```
# Check duplicates
print(df.duplicated().sum())

# Remove duplicates
df = df.drop_duplicates()
```

Purpose

- Remove repeated data
- Improve accuracy of machine learning models

Step 6: Handle Categorical Data

Categorical data was converted into numerical values using Label Encoding.

Code

```
from sklearn.preprocessing import LabelEncoder

label_encoder = LabelEncoder()

for col in categorical_cols:
df[col] = label_encoder.fit_transform(df[col])
```

Purpose

- Convert text values into numbers
- Prepare data for machine learning algorithms

Step 7: Feature Scaling

Numerical columns were standardized using StandardScaler.

Code

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

scaled_features = scaler.fit_transform(df[numerical_cols])
```

Purpose

- Normalize numerical values
- Improve machine learning model performance
- Reduce differences between large and small values

Step 8: Save Cleaned Dataset

The cleaned dataset was saved as a new CSV file.

Code

```
# Save cleaned dataset
df.to_csv('cleaned_dataset.csv', index=False)
```

Purpose

- Store processed data
- Use cleaned dataset in future machine learning tasks

Final Outcome

After preprocessing:

- Missing values were handled
- Duplicate rows were removed
- Categorical data was encoded
- Numerical features were scaled
- Clean dataset was successfully prepared

Conclusion

This project successfully transformed raw data into clean and structured data suitable for machine learning.

Data preprocessing improves dataset quality and helps machine learning models perform better.

Final Workflow

Raw Data → Clean Data → Prepared Data